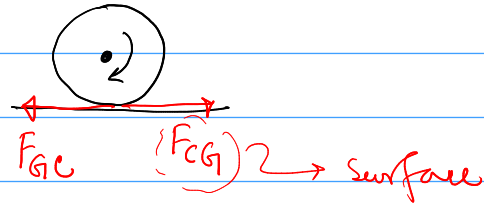
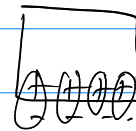
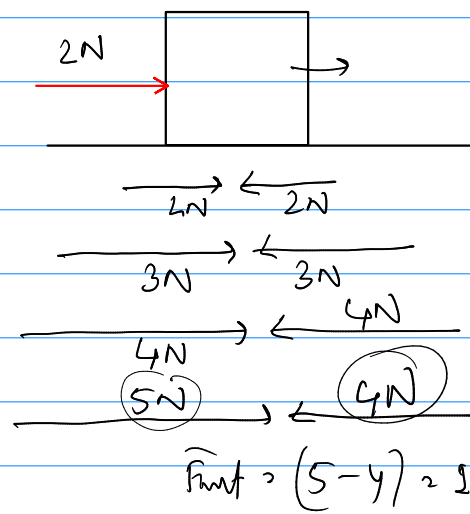


Friction



- ① static frictional force $\rightarrow f_s$
- ② kinetic frictional force $\rightarrow f_k$



$$f_{s, \max} = \mu_s N$$

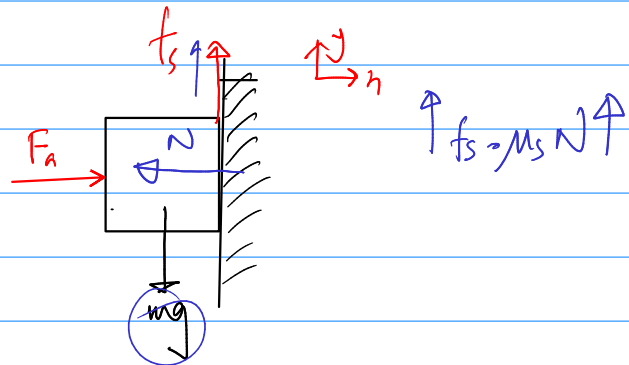
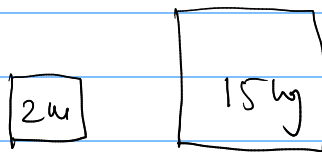
μ_s - coefficient of static friction

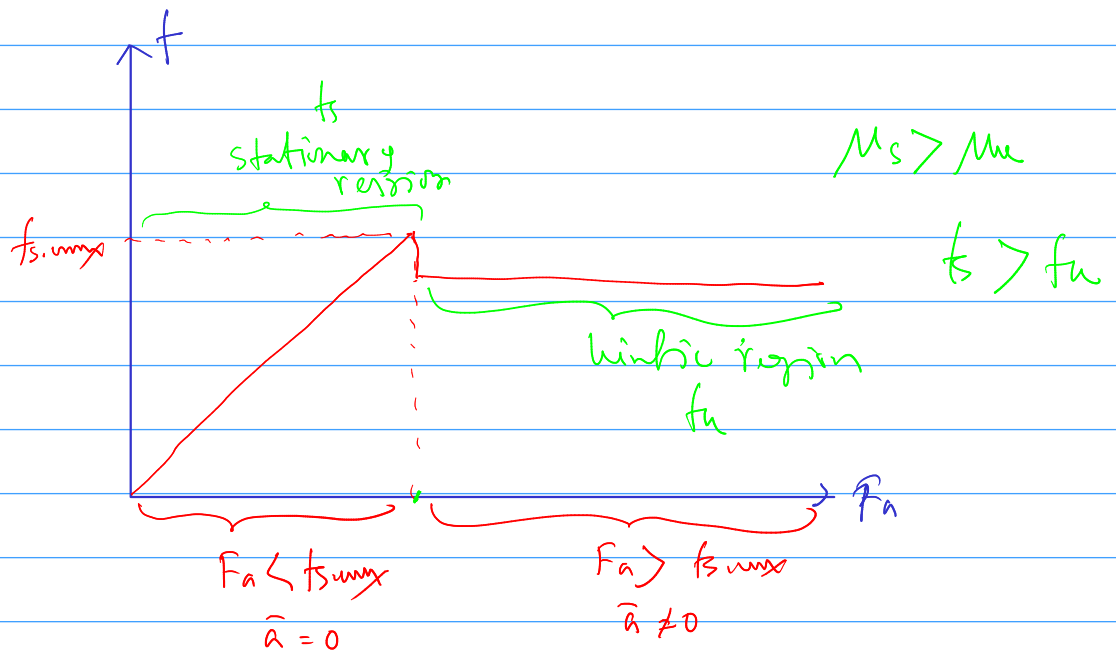
N - Normal reaction force.

$$F_a > f_{s, \max}$$

$$f_k = \mu_k N$$

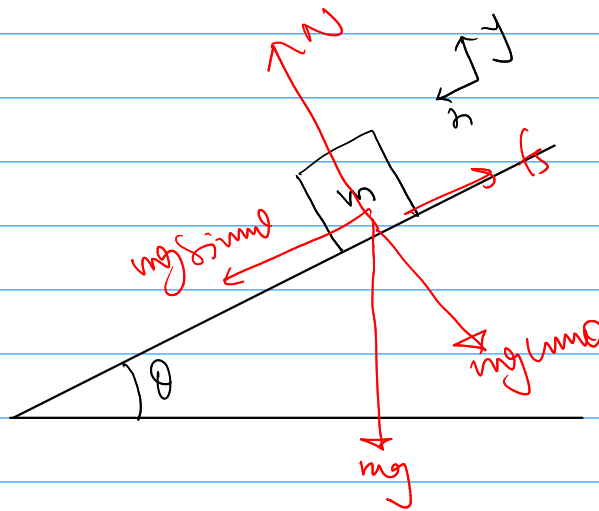
μ_k - coefficient of kinetic friction





$$0 \leq f \leq f_{s, \max}$$

Limiting angle!



$$\sum F_y = 0$$

$$\Rightarrow N - mg \cos \theta = 0$$

$$\Rightarrow mg \cos \theta = N \quad \text{--- (i)}$$

$$mg \sin \theta > f_{s, \max}$$

$$\sum F_x = 0$$

$$mg \sin \theta - f = 0$$

$$\Rightarrow mg \sin \theta = f_s \quad \text{--- (ii)}$$

$$\left[\frac{\text{(ii)}}{\text{(i)}} \right] \Rightarrow$$

$$\frac{mg \sin \theta}{mg \cos \theta} = \frac{f}{N}$$

$$f = \mu_s N$$

$$\Rightarrow \tan \theta = \frac{f}{N} = \mu_s$$

$$\Rightarrow \theta = \tan^{-1}(\mu_s)$$

$$\mu_s = 0.5, \mu_k = 0.34$$

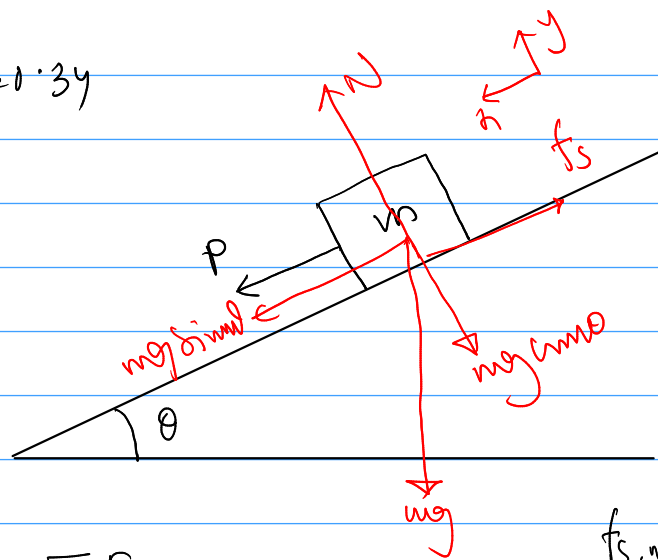
$$mg = 45 \text{ N}$$

$$\theta = 15^\circ$$

$$\vec{P} = 5 \text{ N}, \\ 8 \text{ N}, \\ 15 \text{ N}$$

$$\textcircled{i} \vec{a} = ?$$

$$\textcircled{ii} f = ?$$



$$\sum F_y = 0$$

$$\Rightarrow N - mg \cos \theta = 0$$

$$\Rightarrow N = mg \cos \theta = 45 \cos 15^\circ = 43.5 \text{ N}$$

$$f_{s, \max} = \mu_s N$$

$$f_{s, \max} = 0.5 \times 43.5$$

$$f_{s, \max} = 21.7 \text{ N}$$

$$\textcircled{i} \text{ If } P = 5 \text{ N},$$

$$F_d = P + mg \sin 15^\circ$$

$$= 5 + 45 \sin 15^\circ$$

$$= 16.65 \text{ N} < f_{s, \max}$$

$$\vec{a} = 0, \quad f_s = 16.65 \text{ N}$$

$$\textcircled{ii} \text{ If } P = 8 \text{ N},$$

$$F_d = P + mg \sin 15^\circ = 8 + 45 \sin 15^\circ$$

$$F_d = 19.64 < f_{s, \max}$$

$$\vec{a} = 0, \quad f_s = 19.64 \text{ N}$$

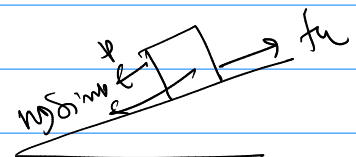
$$\textcircled{iii} \text{ If } P = 15 \text{ N},$$

$$F_d = 15 + 45 \sin 15^\circ = 26.64 \text{ N} > f_{s, \max}$$

$$\sum F_{\text{net}} = ma$$

$$\Rightarrow mg \sin 15^\circ + P - f_k = ma$$

$$\Rightarrow mg \sin 15^\circ + P - \mu_k N = ma$$

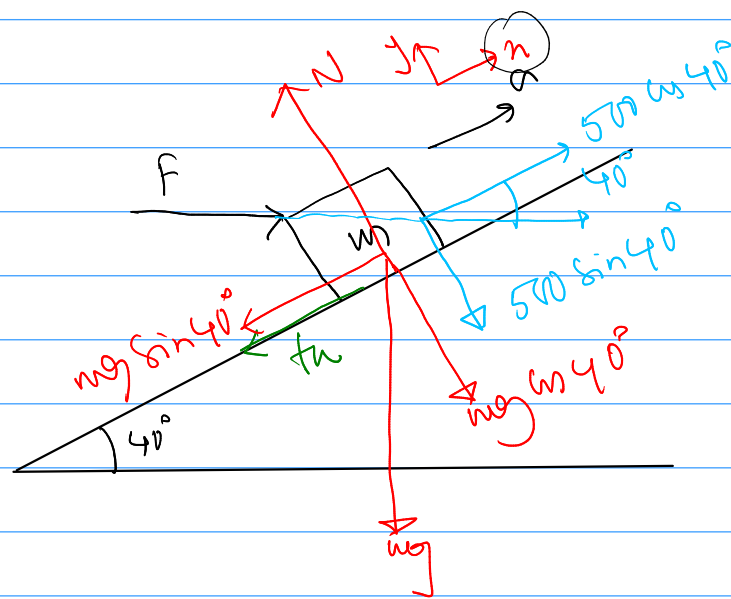


$$\Rightarrow a = \frac{mg \sin 15^\circ + P - \mu_k N}{m}$$

$$\Rightarrow \boxed{\vec{a} = 2.54 \hat{i} \text{ m/s}^2} \quad \underline{\text{Ans}}$$

$$f_k = \mu_k N = 0.34 \times 43.5 = 14.8$$

$$\boxed{\vec{f}_k = -14.8 \hat{i} \text{ N}} \quad \underline{\text{Ans}}$$



$$m = 25 \text{ kg}$$

$$F = 500 \text{ N}$$

$$a = 0.75 \text{ m/s}^2$$

$$\mu_k = ?$$

$$\sum F_y = 0$$

$$\Rightarrow N - mg \cos 40^\circ - 500 \sin 40^\circ = 0$$

$$\Rightarrow N = mg \cos 40^\circ + 500 \sin 40^\circ$$

$$\Rightarrow \boxed{N = 509.7 \text{ N}}$$

$$\sum F_x = ma$$

$$\Rightarrow 500 \cos 40^\circ - mg \sin 40^\circ - f_k = ma$$

$$\Rightarrow 500 \cos 40^\circ - mg \sin 40^\circ - \mu_k N = ma$$

$$\Rightarrow \mu_k = \frac{500 \cos 40^\circ - mg \sin 40^\circ - ma}{N}$$

$$\Rightarrow \boxed{\mu_k = 0.40}$$

$$f_s = \mu_s N$$

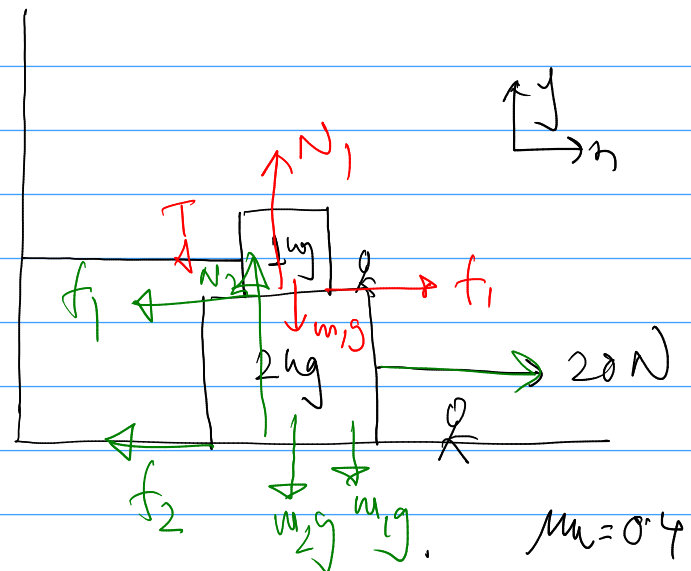
$$f_k = \mu_k N$$

For m_1

$$\sum F_y = 0$$

$$\Rightarrow N_1 - m_1 g = 0$$

$$\Rightarrow N_1 = m_1 g = 9.8 \text{ N}$$



Now, $\sum F_x = 0$

① $T = ?$

$$\Rightarrow T - f_1 = 0$$

② $a = ?$

$$\Rightarrow T = f_1 = \mu N_1 = 0.4 \times 9.8$$

$$\Rightarrow T = 3.92 \text{ N} \quad \underline{\text{Ans}}$$

For m_2

$$\sum F_y = 0$$

$$\Rightarrow N_2 - m_1 g - m_2 g = 0$$

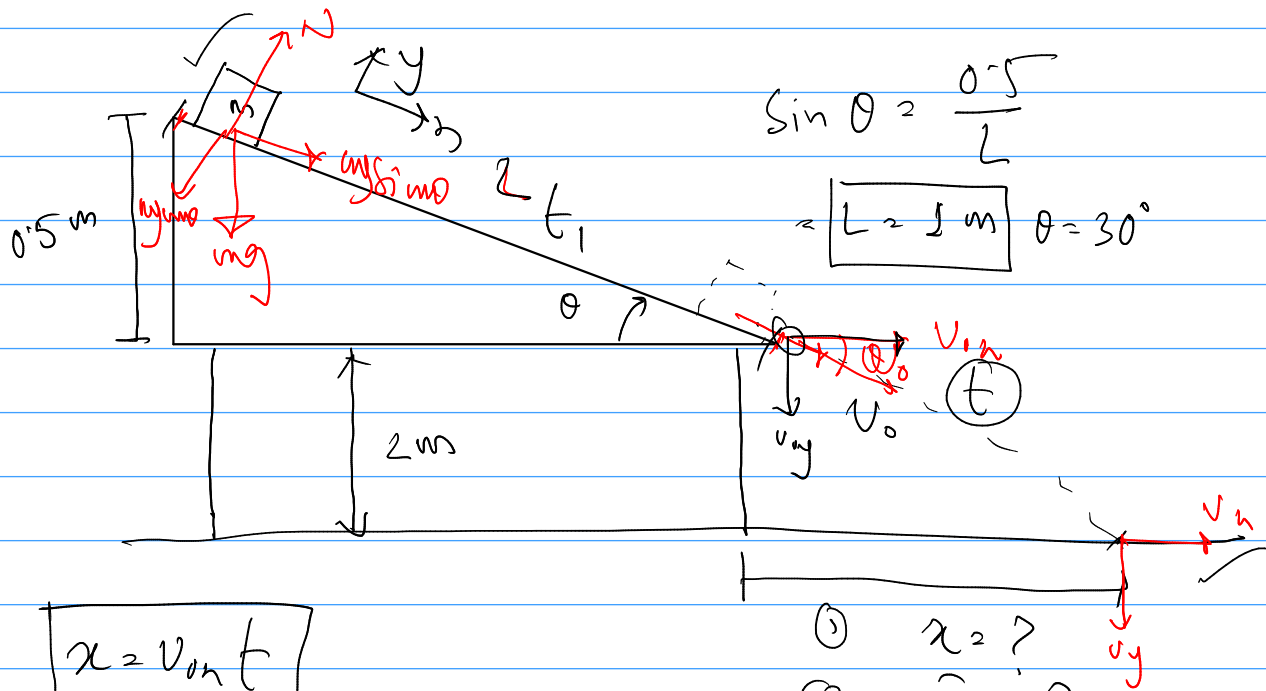
$$\Rightarrow N_2 = (m_1 + m_2)g = 29.4 \text{ N}$$

$$\sum F_x = ma$$

$$\Rightarrow 20 - f_1 - f_2 = m_2 a$$

$$\Rightarrow a = \frac{20 - \mu N_1 - \mu N_2}{m_2}$$

$$\Rightarrow a = 2.16 \text{ m/s}^2 \quad \underline{\text{Ans}}$$



Soln
①

$$x = v_{0x} t$$

$$\sum F_x = ma$$

$$mg \sin \theta = ma$$

$$a = g \sin \theta$$

$$a = 4.9 \text{ m/s}^2$$

$$v^2 = v_{0x}^2 + 2ax$$

$$v = \sqrt{2ax} = \sqrt{2 \times 4.9 \times 1}$$

$$v = 3.12 \text{ m/s} = v_0$$

$$v_{0x} = v_0 \cos \theta = 3.12 \times \cos 30^\circ = 2.7 \text{ m/s}$$

$$y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2$$

$$0 = +2 - 3.12 \sin 30^\circ t - 4.9 t^2$$

$$t = 0.499 \text{ s}, \quad t = -0.82 \text{ s}$$

$$x = v_{rx} \times t = 2.7 \times 0.499 = 1.3 \text{ m} \quad \underline{\text{Ans}}$$

⑩ $\vec{v} = v_x \hat{i} + v_y \hat{j}$

$$v_x = v_{rx} = 3.12 \times \cos 30^\circ = 2.7 \text{ m/s}$$

$$\vec{v} = 2.7 \hat{i} - 6.36 \hat{j}$$

$$v_y = v_{oy} + a_y t$$

Ans $v_y = 3.12 \sin 30^\circ - 9.8 \times 0.499$

$$v_y = -6.36 \text{ m/s}$$

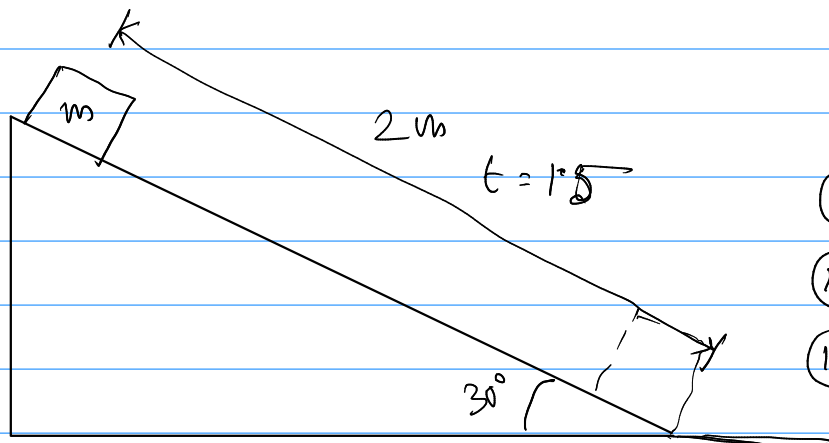
⑭ $T = t_1 + t$

$$v_x = v_{ox} + a_x t$$

$$0 = \frac{v_x}{a_x} = \frac{3.12}{4.9} = 0.637$$

$$T = 0.637 + 0.499 = 1.13 \text{ s}$$

H.W

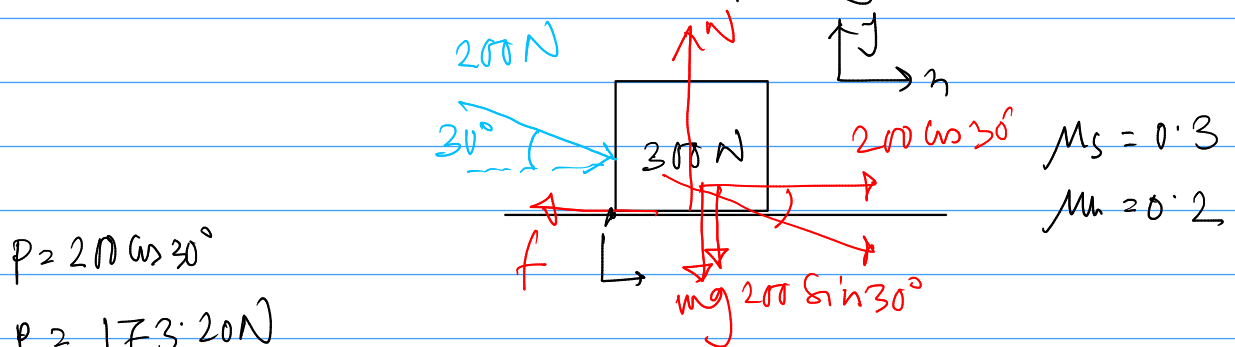


① $a = ?$

② $\mu = ?$

④ v at $t = 1.5$

* Calculate the position and velocity of the block at $t = 5 \text{ sec}$ after applying the force.



$$\sum F_y = 0$$

$$N - mg - 200 \sin 30^\circ = 0$$

$$N = 410 \text{ N}$$

$$f_{s, \max} = \mu_s N$$

$$200 \cos 30^\circ > f_{s, \max}$$

$$f_{s, \max} = \mu_s N = 0.3 \times 410 = 123 \text{ N} < P$$

$$\sum F_x = ma$$

$$200 \cos 30^\circ - f_k = ma$$

$$a = \frac{200 \cos 30^\circ - \mu_k N}{m}$$

$$a = \frac{200 \cos 30^\circ - 0.2 \times 410}{300/9.8}$$

$$\vec{a} = 3.04 \text{ m/s}^2 \hat{i}$$

$$a = \frac{dv}{dt}$$

$$v = v_0 + at$$

$$v = \int a dt = \int a dt$$

$$v(t) = \int a dt + C$$

$$v(t) = \int 3.04 \, dt + C$$

$$v(t) = 3.04t + C$$

$$at = 0$$

$$\underline{\underline{v(0) = 0 + C}}$$

$$\underline{\underline{C = 0}}$$

$$\boxed{v(t) = 3.04t}$$

$$v = \frac{dn}{dt}$$

$$n = \int dn = \int v \, dt$$

$$n(t) = \int (3.04t + C) \, dt + C_1$$

$$n(t) = 3.04 \frac{t^2}{2} + Ct + C_1$$

$$\underline{at \, t = 0 \, \text{Sec}}$$

$$n(t) = 0 = 0 + 0 + C_1$$

$$\boxed{C_1 = 0}$$

$$\boxed{n(t) = 1.52t^2}$$

$$at \, t = 5 \, \text{Sec}$$

$$\bar{v} = 3.04 \times 5 = 15.2 \, \hat{i} \, \text{m/s}$$

$$\bar{n} = 1.52 \times 5^2 = 38 \, \hat{i} \, \text{m}$$

